Abstract

Abstract I

DigiPPPiP extends Partner Pen Play in Parallel (PPPiP) from a co-present paper practice into a reproducible framework for cyberphysical, remote, semisynchronous, asynchronous, accessible, and place-responsive dyadic drawing [@mikhailova2018pppip]. The framework is positioned within collaborative-work studies of shared drawing surfaces [@tang1991collaborativework], cyber-physical-social systems [@sobb2023cpss], mediated embodiment and presence [@biocca1997cyborg;@lee2004presence], digital intimacy technologies [@hassenzahl2012love;@neustaedter2012intimacy], and cautious hyperscanning methodology [@czeszumski2020hyperscanning;@hamilton2021hyperscanning].

Abstract II

The manuscript translates the project brief into a modular research artifact: 9 temporal-spatial modalities, 13 evidence dimensions, 6 planned outcome measures, and 39 conceptual figures are generated or registered by the project code rather than hand-maintained prose. The computational layer is explicitly illustrative, not empirical: the active-inference, inter-brain-synchrony, Forman-Ricci, narrative-information, source-quality, accessibility, and neuroergonomic outputs are deterministic conceptual models designed to make theoretical commitments inspectable. Formal equations are consolidated in [@ sec:formalisms_appendix] so the main line can state evidence boundaries before presenting mathematical detail.

Abstract III

The argument is that DigiPPPiP should be treated as human-human relational technology. Its irreducible design kernel is modest: two partners, a shared mark field, perceptible traces of agency, a temporal relation among contributions, and consentful control over persistence. Digital infrastructure expands the substrate of shared mark-making without replacing the second-person, embodied, narrative, and affective conditions that make the practice relationally meaningful [@schilbach2013secondperson;@bolis2024secondperson;@vaisvaser2024neurodynamics]. The manuscript explicitly avoids claims that DigiPPPiP is clinically therapeutic, universally accessible, or causally validated by neural synchrony; those remain empirical questions for controlled studies. Cross-references use Pandoc labels, so sections, equations, figures, and tables remain automatically numbered across PDF and web render targets.